CareerAI

Analyze • Match • Grow

TEAM MEMBERS
Namrata • Bhagya • Sagarika • Sarika • Sirisha

LEARNING GROUP
LG - 09

AI-POWERED

Resume Analysis & Career Guidance System

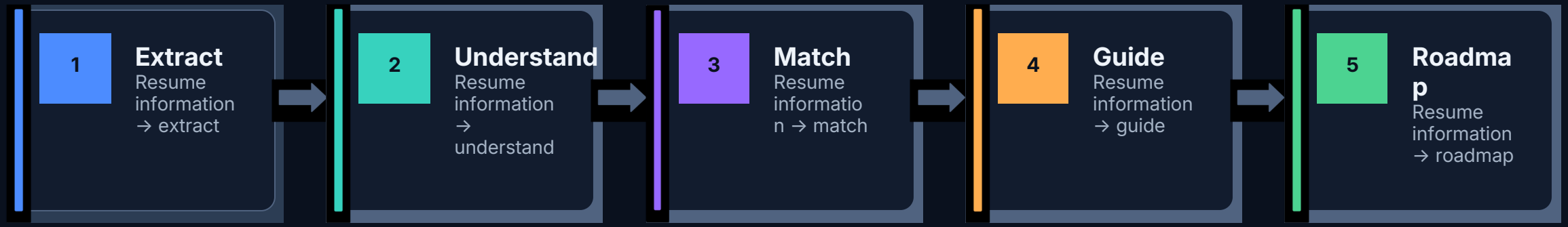
Smarter Insights. Better Choices. A Brighter Future.



02

Abstract

The system turns an uploaded resume into structured career insights by combining PDF extraction, NLP, machine learning and generative AI.



Core workflow

- PDF text extraction and preprocessing
- Skill identification and career matching
- Skill-gap analysis and AI-generated guidance
- Personalized roadmap and consolidated outputs

Made with GAMMA

Introduction

A resume contains evidence of what a person has learned and built — but it does not automatically explain what to do next.



PROJECT IDEA

Resume → Understand → Match → Guide → Plan

04

Problem Statement

Students have information, but lack a connected way to convert it into practical career direction.

01 **Skills are unclear**
Which skills are strongest?

02 **Career choices**
Which roles align?

03 **Skill gaps**
What is missing?

04 **Next steps**
What should I learn?

CONNECTED

05

Existing System

Traditional approaches are fragmented experiences — the motivation for this project.

1. **Manual Resume Review**
   Separate tools and decisions
2. **Career Websites**
   Separate tools and decisions
3. **Skill Assessment**
   Separate tools and decisions
4. **Learning Platforms**
   Separate tools and decisions

FRAGMENTED EXPERIENCE

Resume → Manual Review → Generic Advice → Separate Tools → No single connected career journey

06

Proposed System



ANALYZE
Understand the resume

M ATC H
Find related careers

GAPS
Identify missing skills

GUIDE
Generate AI guidance

PLAN
Build roadmap

ONE CONNECTED CAREER WORKFLOW

Resume → Evidence → Similarity → Insight → Action

Project Objectives

Build an intelligent system that analyzes resumes, matches careers and provides personalized guidance for better career decisions.

Project Objectives

ANALYZE • MATCH • GUIDE • GROW



Resume Analysis 01

Analyze uploaded resume content and structure.

Skill Extraction 02

Identify relevant technical and professional skills.

Career Matching 03

Match the candidate profile with suitable career paths.

Skill Gap Analysis 04

Identify skills that may need further development.

Personalized AI Guidance 05

Generate career-oriented recommendations using AI.

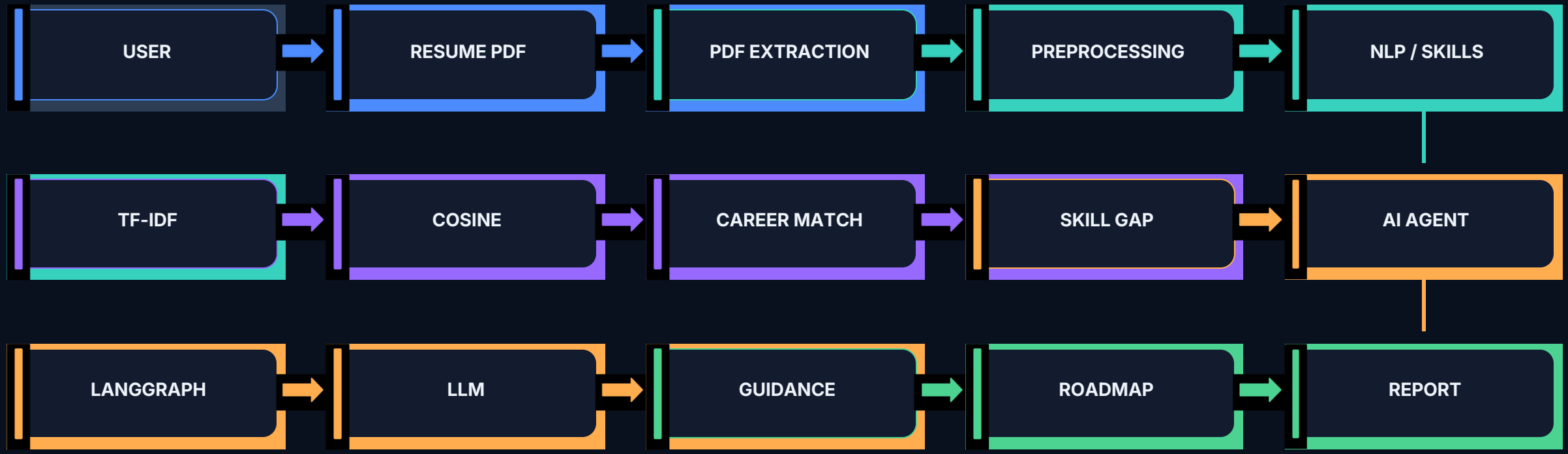
Career Roadmap 06

Provide a structured path for future skill development.

08

Project Architecture

Application layers and the end-to-end technical backbone.



09

Technology Stack

FRONTEND

- React
- HTML
- CSS
- JavaScript

BACKEND

- Py thon
- FastAPI

ML / NLP

- TF-IDF
- Cosine Similarity
- Text Processing

AI

- AI Agent
- LangGraph
- LLM

DEVELOPMENT

- VS Code
- Git / GitHub

The stack separates interface, API, analytical processing and generative guidance.

Major System Components

1. **Resume Upload**
   Accepts the user's resume
2. **PDF Extraction**
   Converts PDF content into text
3. **Preprocessing**
   Cleans and organizes extracted text
4. **Skill Extraction**
   Identifies relevant skills and terms
5. **TF-IDF**
   Converts text to numerical representations
6. **Cosine Similarity**
   Measures similarity between vectors
7. **Career Matching**
   Relates resume content to career profiles
8. **Skill Gap**
   Compares current and required skills
9. **AI Agent**
   Coordinates personalized guidance
10. **LLM**
    Converts analysis to natural-language guidance
11. **LangGraph**
    Organizes the AI workflow
12. **FastAPI**
    Connects the frontend and backend

Resume Upload & PDF Extraction

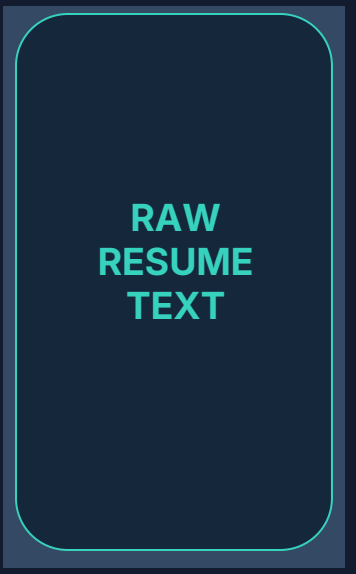
1 UPLOAD PDF

The user provides their own resume.



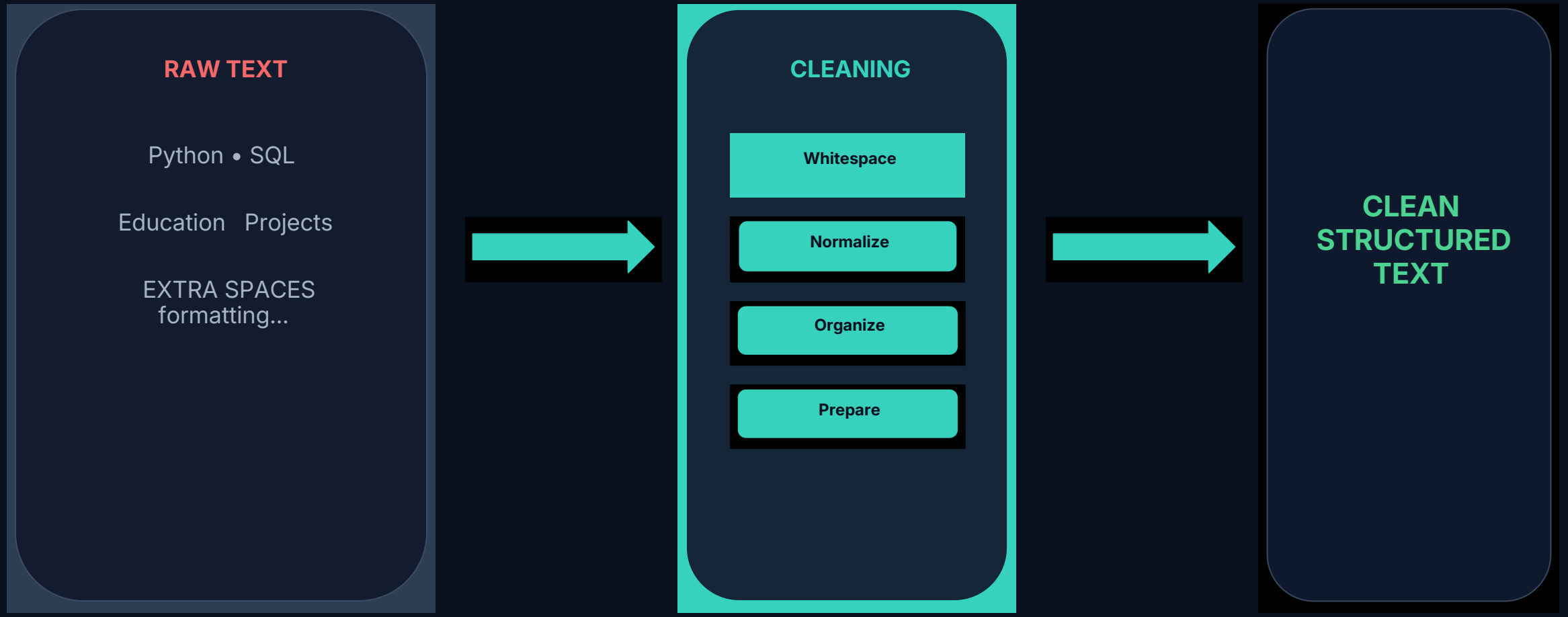
2 EXTRACT TEXT

- Education
- Skills
- Projects
- Experience
- Certifications
- Achievements

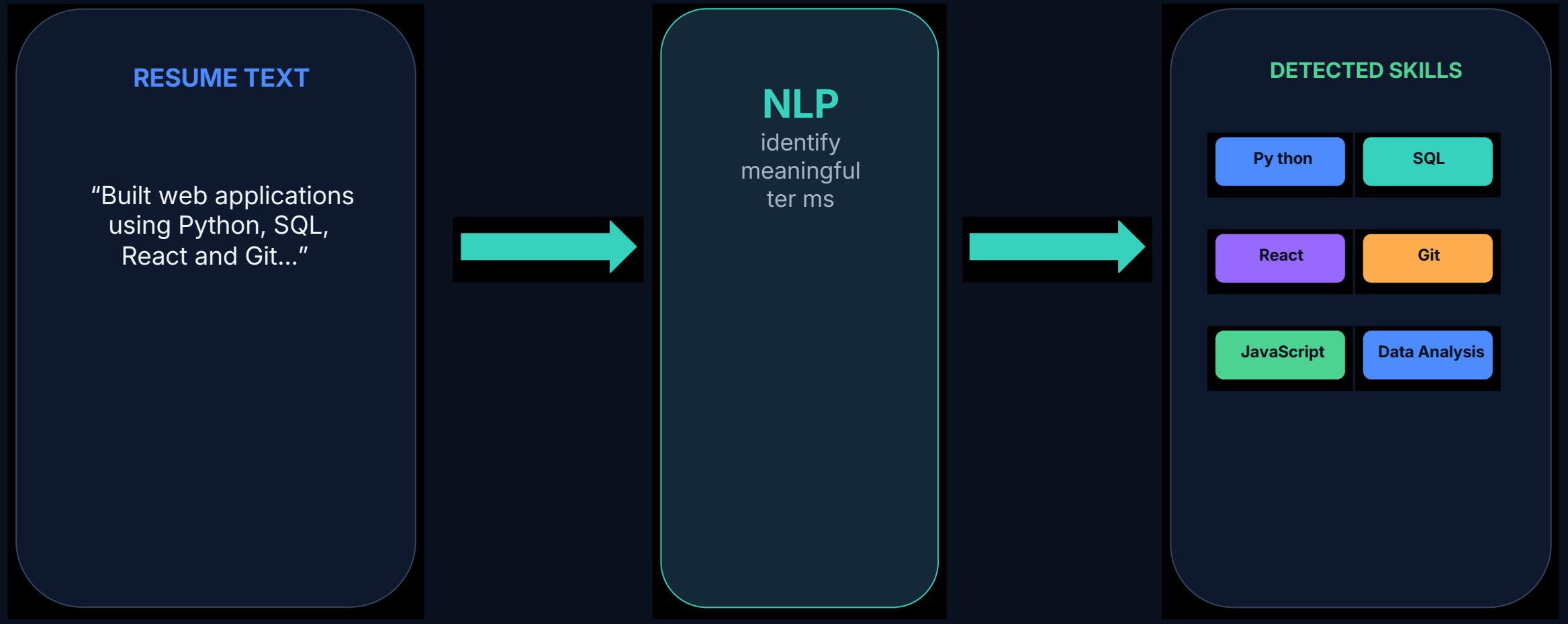


Output: extracted text becomes the input for the next processing stage.

Resume Text Preprocessing



Skill Extraction Using NLP



TF-IDF

Term Frequency–Inverse Document Frequency

TF-IDF converts text into a numerical representation based on term importance.

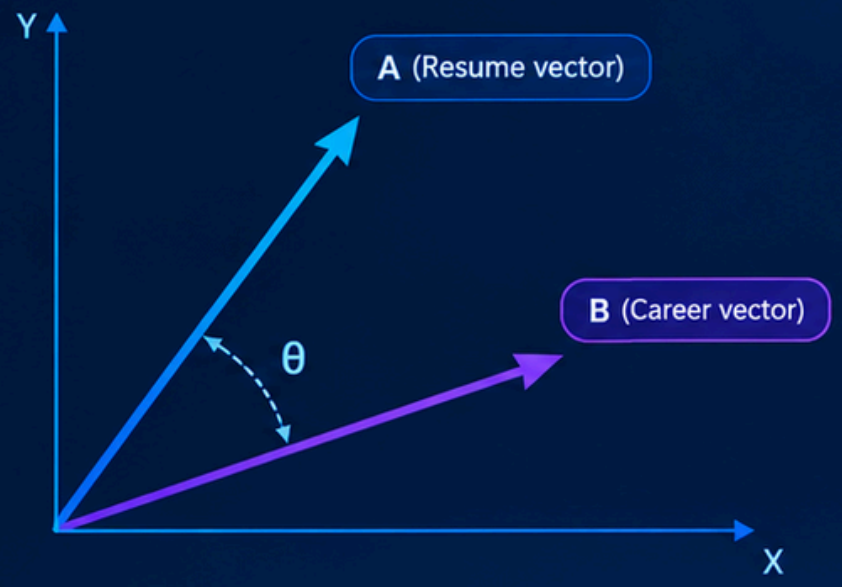


Formula: TF-IDF = TF × IDF

Cosine Similarity

Cosine Similarity compares the direction of two text vectors to estimate their similarity.

Cosine Similarity
COMPARE • MEASURE • MATCH



A — Resume Vector
Numerical representation of resume text.

B — Career Vector
Numerical representation of career profile text.

θ — Angle Between Vectors
Smaller angle = higher similarity.

Formula

$$\text{Cos}(\theta) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

Where,

- $A \cdot B$ → Dot product of the two vectors
- $\|A\|$ → Magnitude of vector A
- $\|B\|$ → Magnitude of vector B

Higher similarity indicates stronger textual similarity between the representations.

Why Use It?
Works well for text data and high-dimensional vectors.

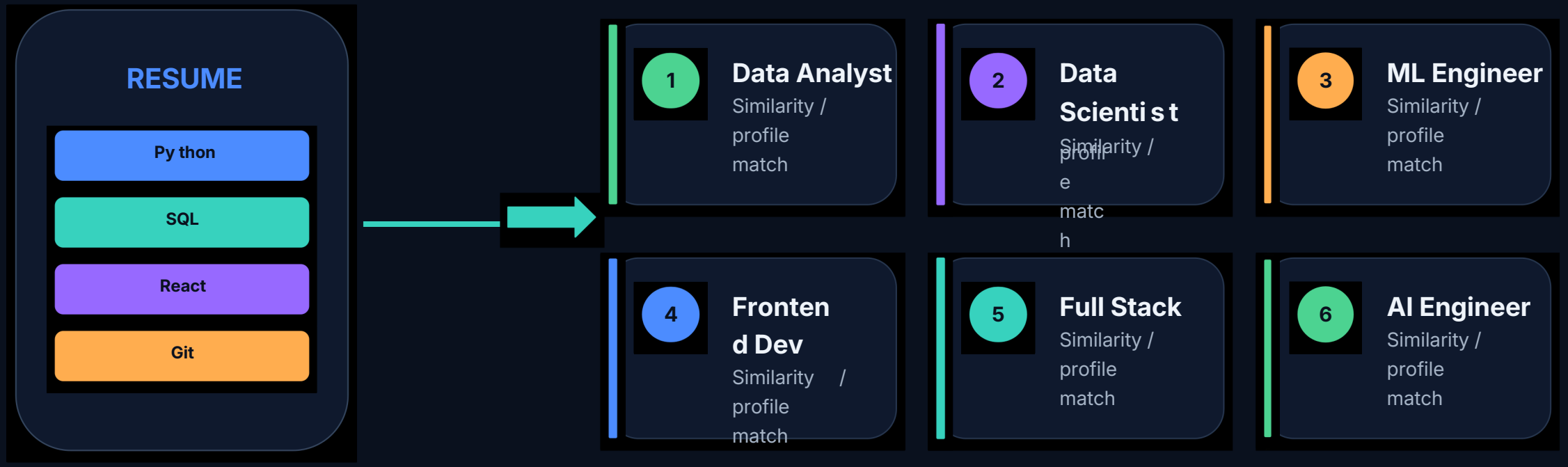
Advantage
Simple, fast and effective for similarity comparison.

Used For
Career matching, recommendation systems, semantic search.

In Our Project
Compares resume and career profile vectors to find the best matches.

Career Matching

The processed resume is compared with available career profiles to identify related roles.



Output: related careers based on resume-to-profile similarity.

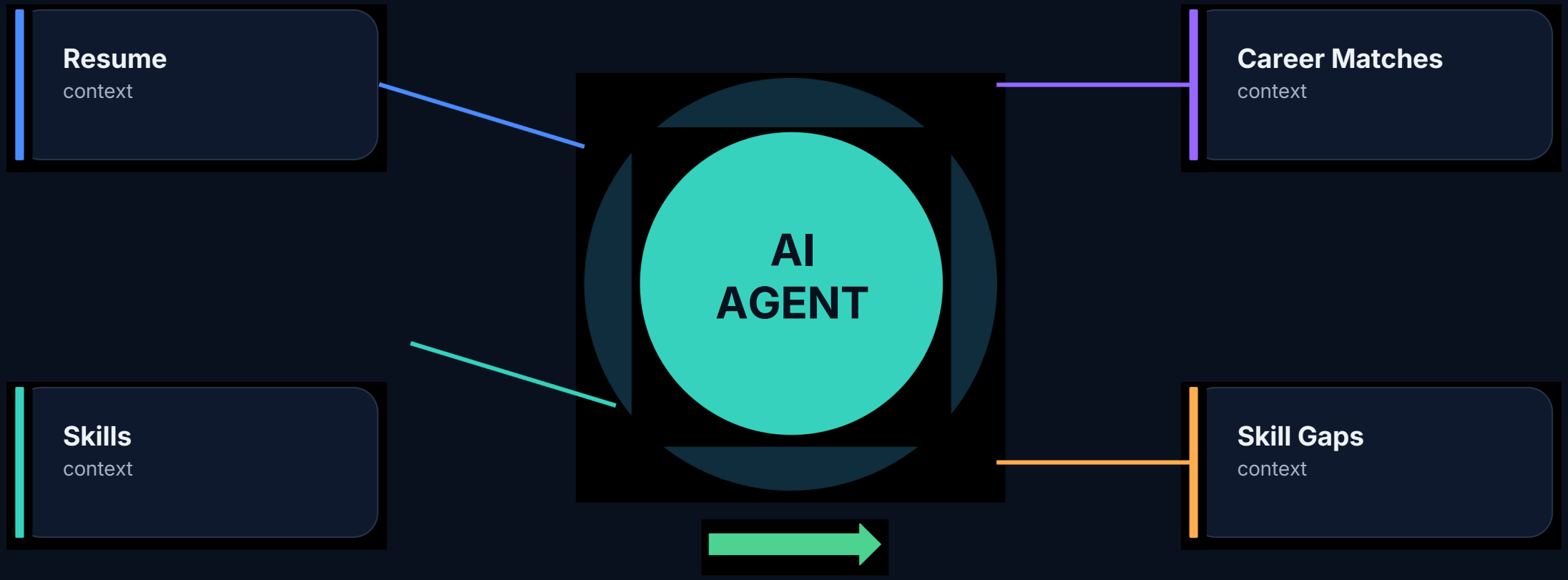
17

Skill Gap Analysis

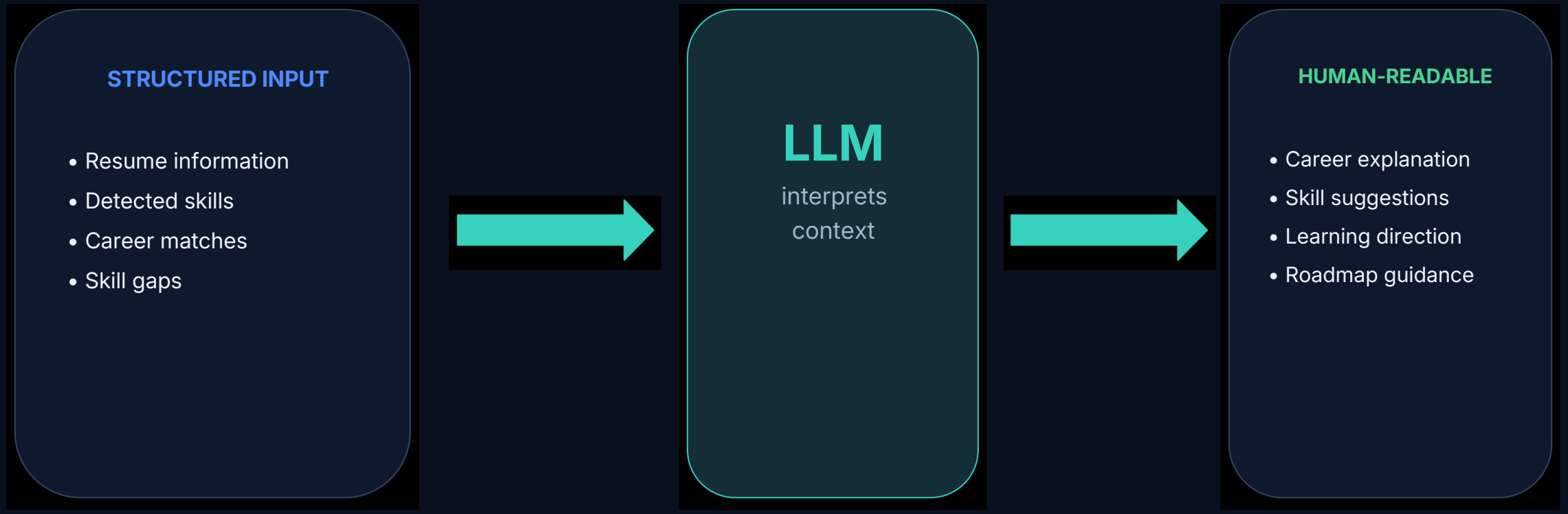


Purpose: compare current skills with the target career and make gaps visible.

AI Career Guidance Agent



Large Language Model (LLM)



Purpose: turn structured analysis into understandable guidance.

LangGraph-Based Workflow

Orchestrates the workflow from resume processing and analysis to **career matching**, skill-gap identification, AI-powered guidance, and roadmap generation.

LangGraph
BUILD · ORCHESTRATE · GUIDE



1 Resume
- Upload PDF/DOCX
- Validate file
- Read content

2 Processing
- Extract text
- Remove noise
- Normalize format
- Structure data

3 ML / NLP Analysis
- Tokenization
- Stopword removal
- Skill identification
- Feature extraction

4 Career Matching
- TF-IDF vectors
- Cosine similarity
- Match profiles
- Rank careers

5 Skill Gap Analysis
- Compare skills
- Identify gaps
- Suggest areas
- Prioritize learning

6 AI Agent
- Orchestrates flow
- Calls tools & models
- Manages context
- Handles logic

7 LLM Reasoning
- Analyzes results
- Generates insights
- Creates suggestions
- Formats response

8 Career Guidance
- Career explanation
- Skill recommendations
- Learning resources
- Personalized advice

9 Career Roadmap
- Learning plan
- Timeline
- Key milestones
- Action steps

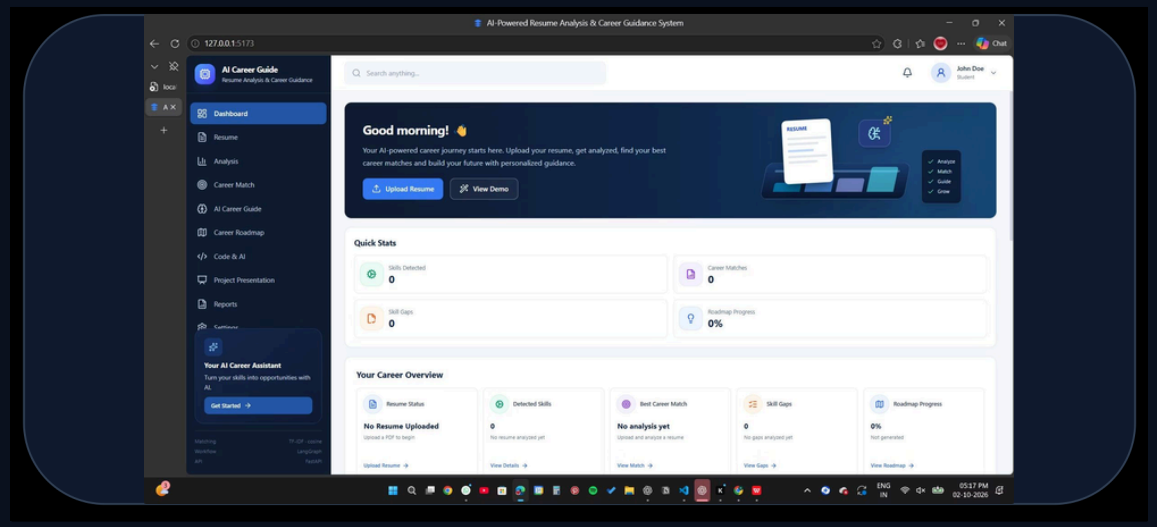
Backend & API — FastAPI

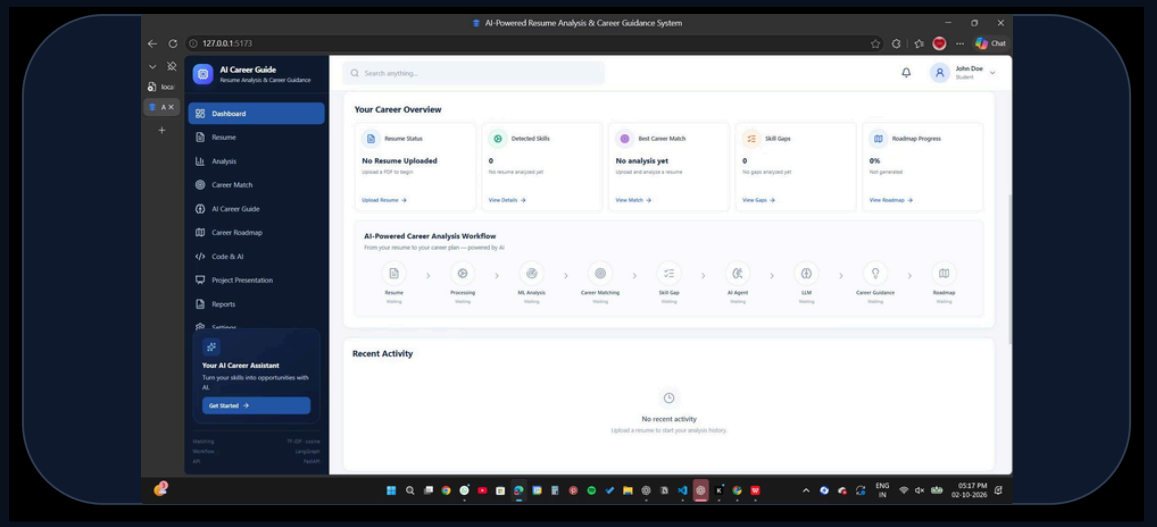


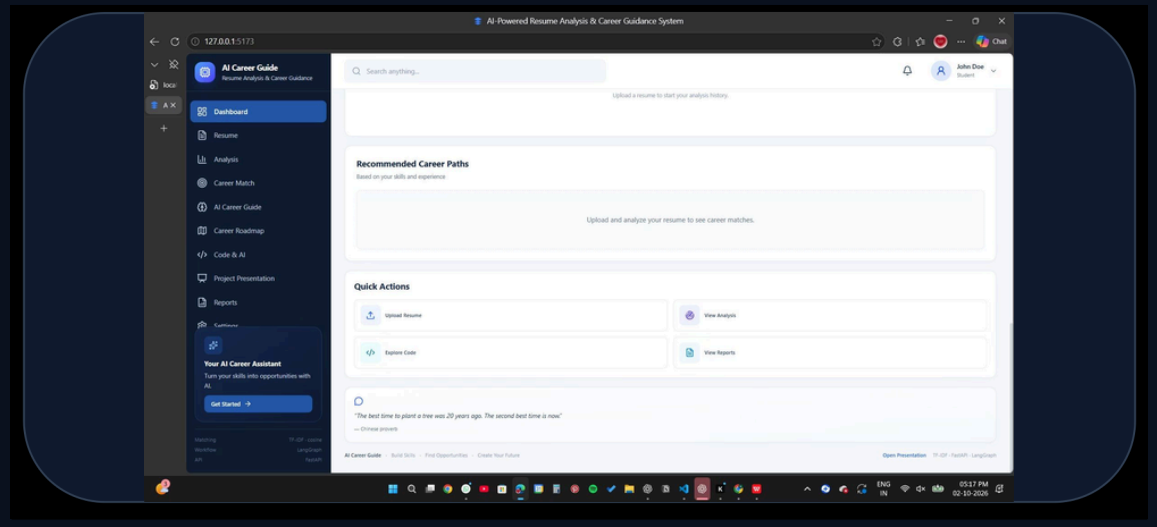
FastAPI provides the communication layer between the interface and Python AI/ML processing services.

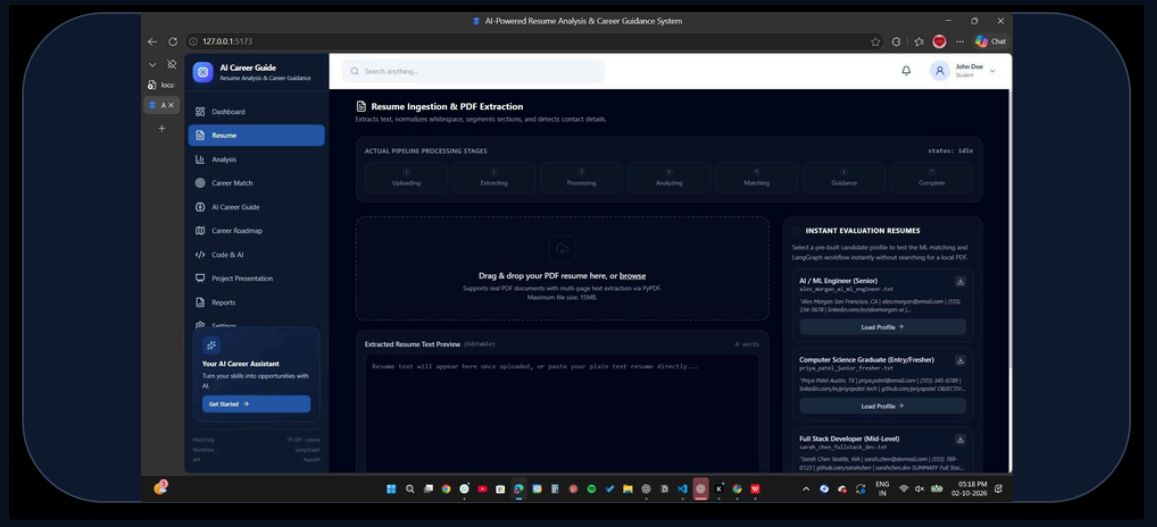
User Interface

Actual implemented project screens — preserved for evaluation.









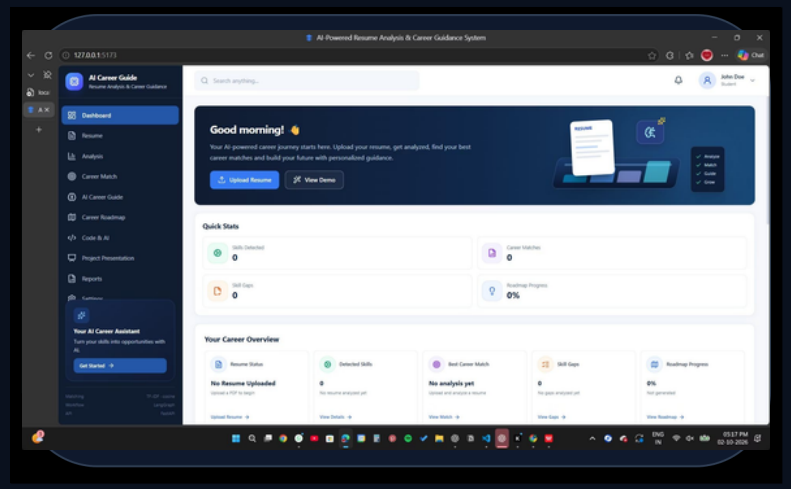
Dashboard / Resume / Analysis / Career workflow

Complete System Workflow

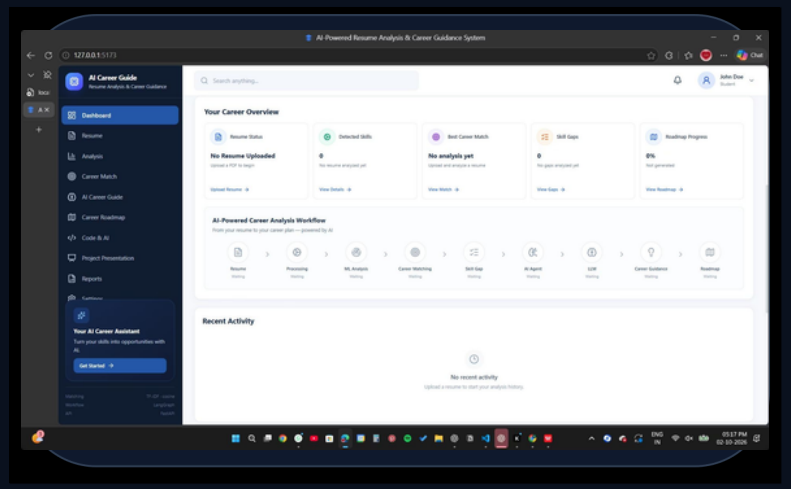


One continuous journey from user input to actionable career output.

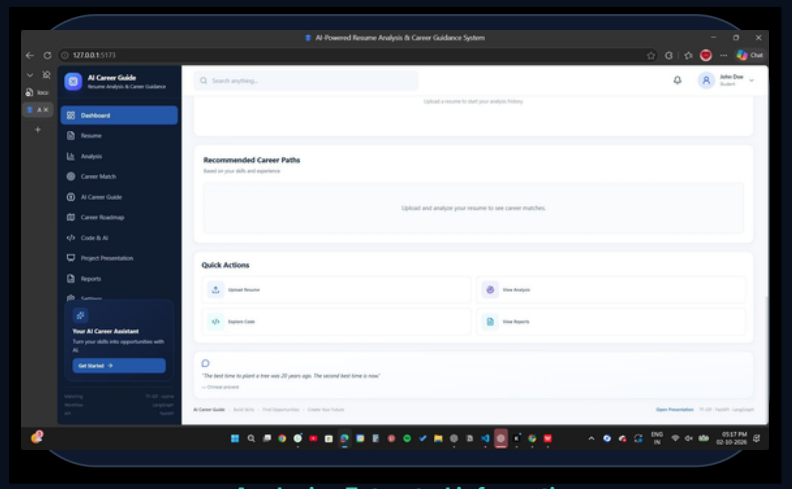
ACTUAL SYSTEM OUTPUTS



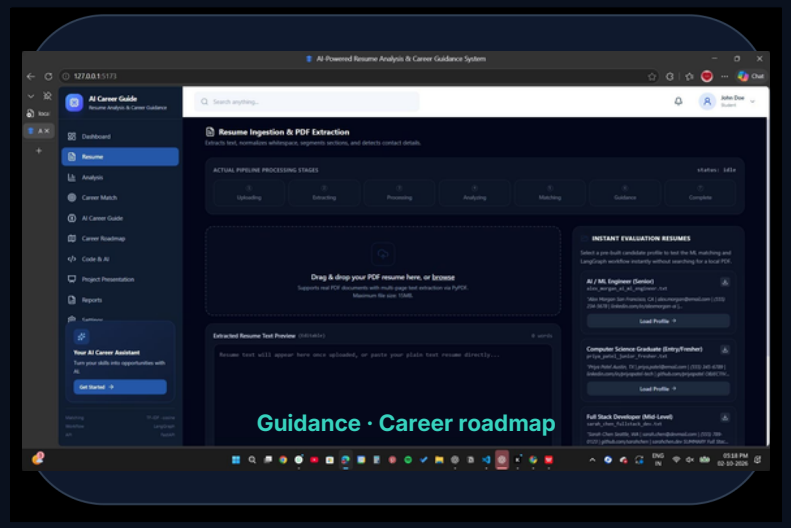
Dashboard · Resume + career status



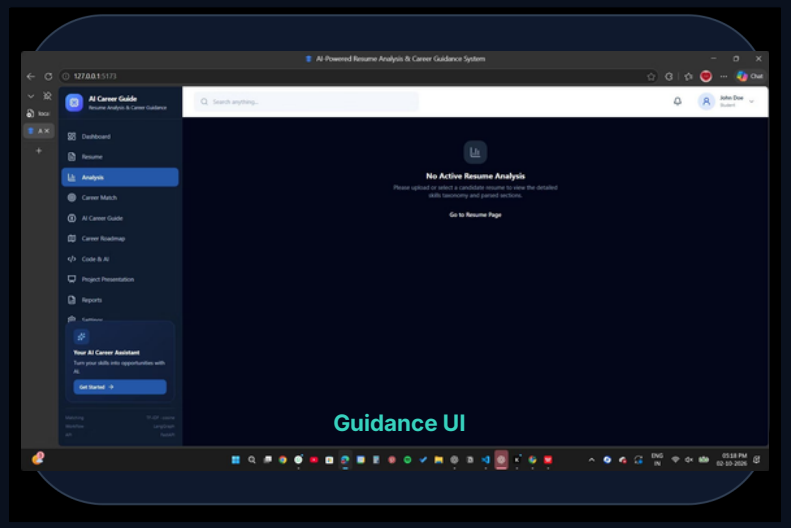
Upload · Process resume



Analysis · Extracted information



Guidance · Career roadmap



Guidance UI

SYSTEM TESTING & VALIDATION

Resume Upload
Input: Resume file
Expected: Resume accepted
Status: Tested in prototype

PDF Extraction
Input: Uploaded resume
Expected: Text extracted
Status: Tested in prototype

Skill Detection
Input: Resume text
Expected: Relevant skills identified
Status: Tested in prototype

Career Matching
Input: Resume + career profiles
Expected: Related career profiles
Status: Tested in prototype

Skill Gap
Input: Skills + target career
Expected: Missing skills identified
Status: Tested in prototype

AI Guidance
Input: Structured analysis
Expected: Career guidance generated
Status: Tested in prototype

Limitations

PDF quality
Poorly formatted or complex PDFs may affect text extraction.

Resume coverage
Analysis depends on the information available in the uploaded resume.

Career profile coverage
Career matching depends on the available career profiles.

Similarity limitations
Text similarity does not represent every aspect of career suitability.

LLM guidance
Generated guidance may require human verification.

Domain coverage
The current system can be expanded to support more career domains.

Decision support
Provides guidance and does not replace human career judgment.

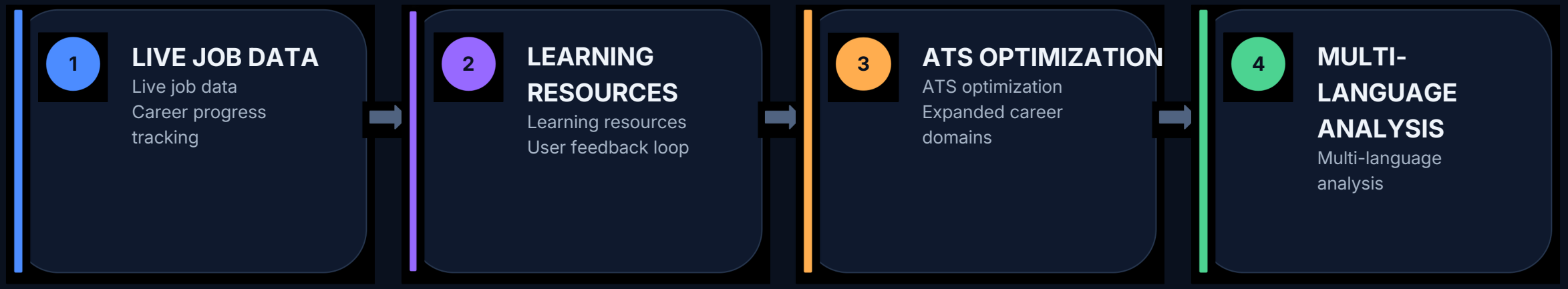
Project scope
The implementation focuses on the defined resume-analysis workflow.

Resume → Guidance → Learning → Jobs → Continuous Career Growth

27

Future Enhancements

Planned extensions can continue the journey from resume guidance to learning, jobs and continuous career growth.



The system is designed as decision-support, not a replacement for human career judgment.

PLANNED EXTENSIONS

Project Value & System Outcomes

The system is designed to make resume information visible, comparable and actionable.



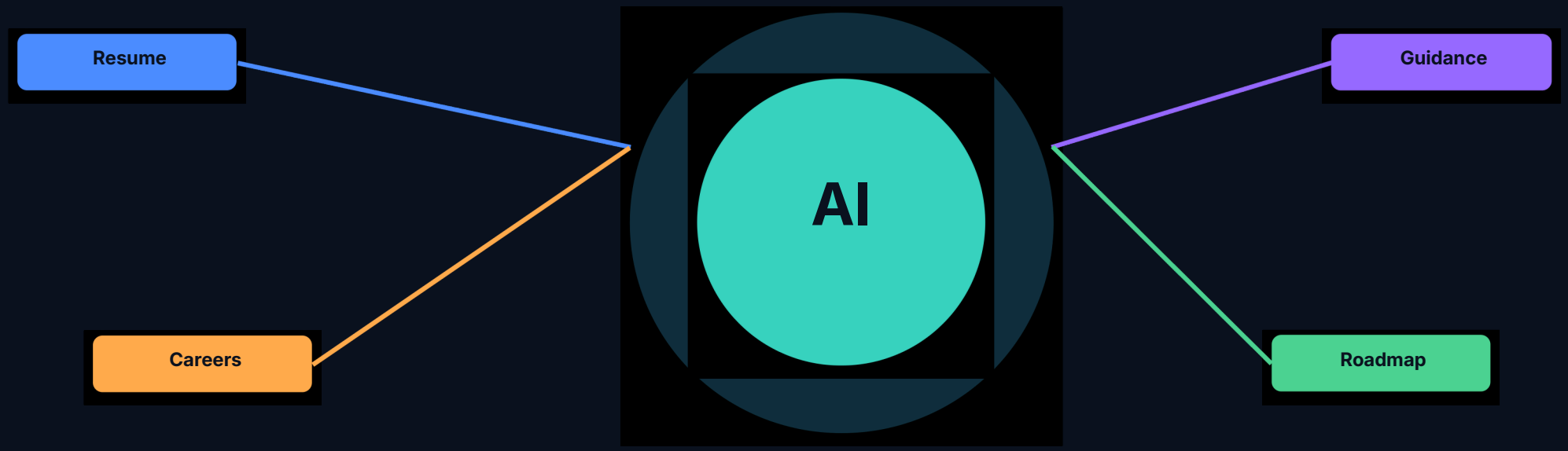
Structured Resume Insights — Converts resume information into organized analysis.
Career Profile Comparison — Relates resume information to available career profiles.
Skill-Gap Identification — Highlights skills that may need further development.
Personalized Guidance — Uses analyzed information to generate career-oriented guidance.

Understand → Compare → Discover → Act

DECISION SUPPORT — NOT A REPLACEMENT FOR HUMAN CAREER JUDGMENT

Conclusion & Key Takeaways

Resume → Skills → Careers → Gaps → Guidance → Roadmap



[TEAM MEMBERS] • [COLLEGE NAME] • [DEPARTMENT] • [ACADEMIC YEAR]

THANK YOU

Questions & Answers

RESUME → CAREER → ROADMAP

